

### Subject Description Form

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| <b>Subject Code</b>                                   | ITC6981RT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Subject Title</b>                                  | Guided Study in Digital Printing in Textile Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Level</b>                                          | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Pre-requisite: Background knowledge in digital printing, knitwear design and knitwear technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Objectives</b>                                     | This subject aims to provide guidance for research students to carry out in-depth study of digital printing and to explore the applications of such technology in creative knitwear design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Intended Learning Outcomes</b>                     | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>assess a range of digital printing technologies critically from a professional perspective.</li> <li>compare the performance and effectiveness of different digital printing programs and devices.</li> <li>experiment and develop a new approach in digital printing on knitted garments.</li> <li>extend the knowledge, techniques and skills in digital printing in knitwear design in order to gain sustainable competitive advantages in the marketplace.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ol style="list-style-type: none"> <li>Mechanism of digital printing on knitwear garments<br/>The principle of digital printing and the operation of such technology in knitwear production</li> <li>Study of digital printing machines and software<br/>A comprehensive research on current digital printing devices and operation programs, the application in relation to knitwear design, comparison of their performance and effectiveness</li> <li>Comparison of visual performance among different materials used in digital printing<br/>Conduct experiments of digital printing on different materials, subject and objective evaluation of visual effect of printed knitwear garments</li> <li>Preparation and finishing in digital printing<br/>The pre-treatment process and procedure before digital printing, the finishing treatment and washing after digital printing</li> <li>Design process and development in digital printing<br/>From design concept to production of knitted fashion items with digital printing, demonstrate critical awareness of latest digital printing technologies and technical innovations, and integrate into the process of fashion design and development</li> <li>Colour calibrating process in digital printing<br/>Analysis of colour performance (in term of vividness, hue, saturation, etc.) if the printed product and adjustment of values of colour input</li> <li>Application of digital printing techniques in creative knitwear design<br/>Design stimulation with computerized devices, generation of creative prototype to accomplish digital printing in knitwear design</li> <li>Sustainability and maintenance of digital-printed knitwear</li> </ol> |

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|                                                                 | Discussion and analysis on aging issue and colour vividness of printed knitwear garment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                |   |          |   |  |  |
| Teaching/Learning Methodology                                   | Coursework can help students extend their knowledge of digital printing in knitwear design and relate the creations to artistic and commercial application, while examination can testify students’ understand of this subject by presentation and submission of written report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |                                                                                |   |          |   |  |  |
| Assessment Methods in Alignment with Intended Learning Outcomes | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |          |   |  |  |
|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             | a                                                                              | b | c        | d |  |  |
|                                                                 | 1.Coursework                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 40%         | ✓                                                                              | ✓ | ✓        | ✓ |  |  |
|                                                                 | 2. Examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 60%         | ✓                                                                              | ✓ | ✓        | ✓ |  |  |
|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |                                                                                |   |          |   |  |  |
|                                                                 | Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 100 %       |                                                                                |   |          |   |  |  |
|                                                                 | The materials submitted for this assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ won work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions                                                                                                                                                                                                                                                                                                                                                                              |             |                                                                                |   |          |   |  |  |
| Student Study Effort Expected                                   | Class contact:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                                                |   |          |   |  |  |
|                                                                 | ▪ Tutorials/Discussions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |                                                                                |   | 26 Hrs.  |   |  |  |
|                                                                 | Other student study effort:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |                                                                                |   |          |   |  |  |
|                                                                 | ▪ Presentation of Final work                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |                                                                                |   | 3 Hrs.   |   |  |  |
|                                                                 | ▪ Self-studies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |                                                                                |   | 91 Hrs.  |   |  |  |
|                                                                 | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |                                                                                |   | 120 Hrs. |   |  |  |
| Reading List and References                                     | 1. Botros, W. (2021). <i>Color management and reproduction in digital printing: A vendor neutral guide for understanding colour printing and matching in images, greys, and solid colours</i> . Wissam Botros.<br>2. Carden, S., & Welters, L. (2015). <i>Digital textile printing (textiles that changed the world)</i> . Bloomsbury USA Academic.<br>3. Gligorijevic, V. (2016). <i>Technology of knitting: Theoretical and Experimental Analysis</i> . Lap Lambert Academic Publishing.<br>4. Irwin, K. (2015). <i>Surface design for fabric</i> . Fairchild Books.<br>5. Nickelson, J. (2017). <i>Fine Art Inkjet Printing: The craft and art of the fine digital print</i> . Rocky Nook.<br><br><u>Periodicals:</u><br>1. Fashion and Textiles<br>2. Textile Progress |             |                                                                                |   |          |   |  |  |

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|  | <ul style="list-style-type: none"><li>3. Textile Research Journal</li><li>4. Other relevant journals</li></ul> |
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